SHRI VILE PARLE KELAVANI MANDAL'S

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(Autonomous College Affiliated to the University of Mumbai)

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PROJECT REPORT ON

FinDora — SVKM DJSCE Campus Lost & Found Platform

Cross-Platform Mobile Application Architecture, Material 3 UI, Gemini AI Assistance, REST Client Integration and Form Validation

TRACK: APPLICATION DEVELOPMENT FRAMEWORK (ADF)

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology (B.Tech) in Information Technology

SUBMITTED BY: [Student Name: e.g. Suruchi Makwana] Roll No: [e.g. A046] SAP ID: [e.g. 60004210045]	UNDER THE GUIDANCE OF: [Project Guide: Faculty Guide Name] [Designation & Department] Dwarkadas J. Sanghvi College of Engg.
DEPARTMENT: Department of Information Technology	ACADEMIC YEAR: [2024–2025 / 2025–2026] Date of Submission: [Date of Submission]

Declaration

I hereby declare that the project report entitled "FinDora — SVKM DJSCE Campus Lost & Found Platform (Application Development Framework Track)" submitted to Dwarkadas J. Sanghvi College of Engineering, affiliated to the University of Mumbai, is an authentic record of original cross-platform mobile software development carried out by me under the supervision and guidance of [Project Guide Name], Department of Information Technology.

I further declare that the content of this project report, including the Flutter widget architecture, Dart asynchronous models, Material 3 theming system, Gemini AI semantic classification services, HTTP REST API client, regular-expression-based form validation, offline seed data resilience, and mobile role-based navigation flows, has not been submitted in part or full to any other University or Institution for the award of any degree or diploma.

All sources of technical literature, official software documentation, open-source Flutter packages, and external application development tools referenced in this work have been properly cited and acknowledged in compliance with academic standards.

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Abstract

Smart mobile devices represent the primary computing and communication medium through which undergraduate engineering students and faculty navigate collegiate life. In high-density campus environments such as Dwarkadas J. Sanghvi College of Engineering (DJSCE), misplaced personal belongings occur dynamically across classrooms, lecture halls, computing laboratories, and campus cafeterias. However, the absence of an immediate, native mobile reporting and verification tool forces students to rely on slow, disorganized word-of-mouth reports or uncoordinated social media stories, causing prolonged delays in recovering misplaced property.

To resolve this mobility bottleneck, this report presents the engineering and architectural implementation of FinDora (Application Development Framework — ADF), a high-performance, cross-platform mobile application developed using Google's Flutter framework and the Dart programming language. The mobile client is designed as a first-class citizen of the FinDora campus ecosystem, communicating seamlessly with a shared Next.js REST API backend while maintaining complete independence of presentation, local state, and offline caching.

The technical architecture of the mobile application incorporates several cutting-edge mobile development paradigms: (1) a clean, multi-layered widget hierarchy organized into Presentation, Service, and Data layers; (2) Material 3 design system implementation supporting dynamic dark and light mode synchronization via a custom `ThemeService` notifier; (3) a 5-tab persistent bottom navigation shell (`BottomNavigationBar`) coordinating Home, Browse, Gemini AI Matcher, Report/Post, and Profile modules; (4) an asynchronous HTTP communication service (`ApiService`) featuring automated network timeout handling, host resolution across desktop and emulator targets, and graceful fallback to offline seed datasets (`SeedData`); (5) Gemini AI assistance (`GeminiService`), providing automatic categorical classification from natural language descriptions and semantic match scoring with localized recovery advisories; (6) robust form validation leveraging `GlobalKey<FormState>`, enforcing strict 4-character Roll Number (`^[A-Za-z][0-9]{3}$`), 12-digit SVKM SAP ID (`^6[0-9]{10,11}$`), SVKM email suffixes (`@svkmmumbai.onmicrosoft.com`), and mandatory image proof for found items; (7) real-time claim status tracking on item detail views (`_existingClaim`) directing students to the

Ground Floor Security Desk (Gate 2); and (8) role-based profile isolation separating student digital ID credentials from the Faculty Security Console.

The mobile application was rigorously verified using the Flutter static analysis tool (`flutter analyze`), achieving zero warnings and zero errors across all Dart source files. Functional verification across 15 real-world mobile test scenarios confirmed 100% compliance with institutional validation rules, live REST communication, and error resilience. FinDora establishes an exemplary standard for intelligent, responsive campus mobile applications.

Keywords: Flutter, Dart, Application Development Framework, Material 3, Gemini AI Assistance, REST API Client, Form Validation, Smart Campus.

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List of Abbreviations

ADF	Application Development Framework
AI	Artificial Intelligence
API	Application Programming Interface
APK	Android Package Kit
DJSCE	Dwarkadas J. Sanghvi College of Engineering
FST	Full Stack Track
HTTP	Hypertext Transfer Protocol
IEEE	Institute of Electrical and Electronics Engineers
JSON	JavaScript Object Notation
M3	Material Design 3
NLP	Natural Language Processing
PII	Personally Identifiable Information
RBAC	Role-Based Access Control
REST	Representational State Transfer
SAP	Systems, Applications, and Products in Data Processing (Institutional Student ID)
SDK	Software Development Kit
SVKM	Shri Vile Parle Kelavani Mandal
TF-IDF	Term Frequency-Inverse Document Frequency
UI	User Interface
URI	Uniform Resource Identifier
URL	Uniform Resource Locator
UX	User Experience
VM	Virtual Machine

Chapter 1: Introduction

Modern educational institutions function as dynamic, highly mobile ecosystems where students and faculty continuously traverse classrooms, computer centers, engineering laboratories, libraries, and cafeteria areas. At Dwarkadas J. Sanghvi College of Engineering (DJSCE), an autonomous engineering college affiliated to the University of Mumbai under the management of Shri Vile Parle Kelavani Mandal (SVKM), thousands of students navigate multi-story academic wings daily.

Throughout these active physical transitions, misplaced personal property is a recurring and stressful occurrence. Essential items—such as institutional smart ID cards, high-value electronics (calculators, laptops, headphones, mobile chargers), textbooks, keys, water bottles, and stationery—are frequently left behind in classrooms (e.g., Room 402), central reading halls (Library 3rd Floor), computing labs (IT Lab 3), and student canteen tables.

While web portals provide robust desktop management, students are overwhelmingly mobile-first users. When an item is lost or discovered between lectures, a student does not typically open a laptop, connect to Wi-Fi, and navigate a desktop browser. Instead, the student immediately reaches for their smartphone. Under existing informal campus conditions, mobile property reporting is severely handicapped:

1. Reliance on Ephemeral Social Channels: Recovered items are photographed and broadcast haphazardly across class WhatsApp groups or ephemeral Instagram stories. These posts lack structured search filters, location tagging, or category indexes, disappearing rapidly under hundreds of subsequent peer messages.

2. Immediate Exposure of Private Data: When students announce found items on informal groups, they often attach unredacted photographs displaying student names, Roll Numbers, and government identification numbers, exposing students to harassment and unauthorized contact.

3. High Friction in Urgent Reporting: Without an optimized mobile interface with intelligent input defaults, capturing item details, selecting categories, and uploading photos between busy lecture schedules presents excessive friction.

4. Absence of Verification Integrity: Anyone viewing a social media broadcast can visit the security desk and falsely claim an item before the rightful owner even becomes aware of its recovery.

To eliminate these critical challenges, FinDora (Application Development Framework — ADF) was engineered as a dedicated, cross-platform mobile application developed using Flutter and Dart. Designed to operate as a native mobile client for the SVKM DJSCE campus community, the mobile app integrates seamlessly with a shared Next.js REST API backend, providing real-time property cataloging, Gemini AI-assisted classification, structured ownership verification, and dynamic Material 3 user interfaces.

1.1 Objectives of the System

The technical objectives of the FinDora Mobile Application are established to fulfill the rigorous requirements of the DJSCE ADF curriculum:

1. High-Performance Cross-Platform Portability: To engineer a single, maintainable Flutter codebase in Dart that compiles natively across mobile platforms (Android, iOS) and desktop environments (Windows desktop), ensuring 60 frames-per-second (FPS) rendering and native hardware responsiveness.

2. Real-Time REST Integration with Shared Backend: To implement an asynchronous HTTP client service (`ApiService`) communicating with the shared Next.js REST API endpoints (`/api/items`, `/api/claims`, `/api/ai/match`), enabling unified, real-time data synchronization between web and mobile users without duplicating backend database logic.

3. Gemini AI Assistance for Mobile Users: To incorporate Google Gemini AI capabilities via `GeminiService`: (a) automatic keyword-based category detection during report creation, reducing manual form entry friction; and (b) a dedicated Gemini AI Semantic Matcher screen (Mini Project I) that evaluates reported items against active campus findings with confidence scores and reasoning badges.

4. Material 3 Design System & Dynamic Theming: To implement an intuitive Material Design 3 interface featuring a persistent 5-tab navigation shell (`BottomNavigationBar`), custom reusable widgets (`ItemCard`, `StatusChip`), and dynamic dark/light theme switching managed via a reactive `ThemeService` notifier.

5. Client-Side Verification and Form Validation Parity: To implement robust form validation using `GlobalKey<FormState>`, enforcing strict 4-character Roll Number (`^[A-Za-z][0-9]{3}$`), 12-digit SVKM SAP ID (`^6[0-9]{10,11}$`), SVKM institutional email suffixes (`@svkmmumbai.onmicrosoft.com`), and mandatory image proof for found items.

6. Offline Resilience and Graceful Fallback: To engineer an offline seed data architecture (`SeedData`) within `ApiService` that ensures the mobile application remains fully navigable, searchable, and demonstrable even during localized server outages or campus network interruptions.

1.2 Scope of the Project

The scope of the FinDora Mobile Application encompasses the complete client-side mobile lifecycle:

1. Target User Personas: The mobile client serves SVKM DJSCE students for on-the-go reporting, catalog discovery, card swiping, peer chat, and claim tracking; and campus faculty security personnel for administrative moderation through the Faculty Security Console.

2. Supported Platforms: The Flutter application is engineered and verified for Windows Desktop (`flutter run -d windows`), Chrome Web (`flutter run -d chrome`), and Android mobile targets via Android SDK.

3. Architectural Separation of Concerns: The mobile application strictly delegates database persistence, password hashing, and master claim approvals to the shared Next.js backend, acting as an intelligent, responsive presentation and client-validation client.

Table 1.1: FinDora ADF Concept Mapping to DJSCE Academic Syllabus

Module	Syllabus Curriculum Topic	Implemented Location & Technical Features in mobile_app/
Module 1	Gemini for Application Developers	GeminiService.suggestCategory() for automated keyword classification; Mini Project I: GeminiMatcherScreen executing semantic item matching against backend /api/ai/match.
Module 2 & 3	Introduction to Flutter and Dart Language	Strongly-typed Dart classes (models/item.dart) with JSON serialization (fromJson/toJson); asynchronous programming with async/await and Future.
Module 4	Working with Flutter (Widgets, Forms, Theming)	Material 3 UI implementation; custom widgets (ItemCard, StatusChip); GlobalKey<FormState> input validation; ThemeService for reactive dark/light mode toggle.
Module 5	Building a Flutter Application (Mini Project II)	Multi-page app with 5-tab BottomNavigationBar shell (Home, Browse, Matcher, Report, Profile); ApiService HTTP client consuming shared REST API; SeedData offline fallback; pull-to-refresh.

Chapter 2: Literature Review

The development of intelligent mobile applications for smart campus environments requires a thorough understanding of mobile user interfaces, on-device data processing, client-server communication paradigms, and recommendation techniques [1], [2]. This chapter reviews foundational recommendation and discovery algorithms, analyzes their mobile application, and contrasts existing paradigms with FinDora's mobile architecture.

2.1 Existing Recommendation Techniques

Recommendation systems aim to reduce information overload by presenting users with items most relevant to their context or preferences [3], [4]. In mobile application development, recommendation techniques must balance algorithm precision against device resource constraints (memory, network bandwidth, battery consumption) [14].

1. Lexical Keyword Matching and Token Space Similarity: Traditional item matching systems rely upon lexical token intersection and vector space models, such as Term Frequency-Inverse Document Frequency (TF-IDF) [5], [6]. In mobile architectures, computing heavy TF-IDF matrices on-device can introduce UI thread jank. Consequently, mobile applications typically implement lightweight tokenization or offload heavy vector computations to cloud REST endpoints [1], [14]. While token-based search is fast, it fails to handle semantic variations (e.g., matching 'cellphone' with 'smartphone') [6].

2. On-Device Rule-Based Categorization: In mobile forms, minimizing user keystrokes is critical for data collection accuracy. Rule-based keyword matching uses deterministic lookup tables to infer item categories from input text [7]. By analyzing regular expressions or substring intersections on the client device, the interface can dynamically suggest categories (e.g., inferring 'Electronics' from 'AirPods') without incurring network latency [2], [7].

3. Content-Based Mobile Filtering: Content-based filtering recommends items by comparing structured attribute vectors (category, location, date, color) [3], [4]. In mobile lost-and-found contexts, content-based matching is uniquely advantageous because it operates independently of historical user interaction matrices, functioning immediately when a new report is filed (solving the 'cold start' problem) [4].

4. Collaborative Filtering and Inherent Mobile Limitations: Collaborative filtering algorithms (user-based or item-based matrix factorization) predict interests from collective user rating histories [3], [9]. However, lost-and-found mobile platforms cannot leverage collaborative

filtering: lost reports represent unique, non-recurring physical instances that are permanently retired upon recovery [1], [3].

2.2 Applications of Recommendation Systems in E-Commerce and Lost-and-Found Platforms

Commercial mobile e-commerce applications (such as Amazon and Flipkart) extensively utilize content-based recommendation trees to present personalized item suggestions, related accessories, and category filters [4], [10]. Mobile e-commerce design emphasizes faceted search chips, swipe gestures, and instant feedback.

Applying these principles to mobile campus lost-and-found applications presents distinct operational constraints [1], [2]. Misplaced items lack standardized barcodes or commercial product catalogs. Instead, reports contain unstructured, subjective natural language descriptions submitted under hurried campus conditions [11]. Furthermore, mobile platforms must prevent fraudulent claims by embedding structured verification dialogs before physical handover [12].

Recent research in smart campus mobile architectures [1], [2], [11] explores mobile reporting apps built with native or hybrid frameworks. However, published systems exhibit notable technical gaps:

• Tight Coupling to Monolithic Backends: Many mobile prototypes hardcode backend database calls directly within UI widgets, violating clean separation of concerns and preventing web-mobile interoperability [1], [2].

• Fragility to Network Outages: Most surveyed apps crash or display blank error screens when campus Wi-Fi is lost, lacking local offline caching or fallback datasets [11].

• Form Input Inconsistency: Several mobile implementations omit strict credential validation, allowing arbitrary student inputs that complicate administrative matching [12].

Comparison with FinDora's Implemented Mobile Architecture: FinDora resolves these issues through a decoupled, multi-layered Flutter architecture [13]:

1. Lightweight Client-Side Gemini Categorization: `GeminiService.suggestCategory()` performs instant, on-device keyword inference across major campus categories (Electronics, ID & Cards, Books & Stationery, Accessories, Keys, Clothing), auto-populating form dropdowns in real time without blocking the UI thread.

2. Asynchronous REST Client with Offline Resilience: `ApiService` communicates asynchronously with the shared Next.js `/api/ai/match` route over HTTP. If network connectivity fails, the app seamlessly falls back to its bundled `SeedData` catalog, ensuring zero crashes and continuous usability.

3. Dedicated Gemini AI Matcher Screen (Mini Project I): Rather than a black-box recommendation list, `GeminiMatcherScreen` provides students with an interactive semantic search tool, displaying confidence percentages, keyword overlap justifications, and localized DJSCE recovery advisories.

Chapter 3: Implementation

The implementation of the FinDora Mobile Application was engineered using Google's Flutter framework (v3.24+ / v3.47) and the Dart programming language. The mobile client provides a native, responsive experience across Android, iOS, and Windows desktop platforms, implementing clean architectural boundaries, Material 3 design standards, asynchronous HTTP networking, and Gemini AI assistance.

3.1 System Architecture and Design

The FinDora mobile application is designed around a three-tier architecture: Presentation Layer, Service Layer, and Data Layer. This structure ensures strict modularity, high code maintainability, and seamless integration with the shared Next.js backend.

1. Presentation Layer (Material 3 Widgets & UI):

• Root Application (`main.dart`): Configures `MaterialApp` with Material 3 theming (`useMaterial3: true`), defining light and dark color schemes, typography, and card themes. The application root is wrapped with `themeNotifier` to achieve instant, reactive theme switching.

• Navigation Shell: Houses a responsive `Scaffold` featuring a persistent 5-tab `BottomNavigationBar` (`HomeScreen`, `BrowseScreen`, `GeminiMatcherScreen`, `PostItemScreen`, `ProfileScreen`), enabling fluid, thumb-friendly navigation across primary modules.

• Reusable Widget Library: Custom widgets encapsulated in `lib/widgets/`: `ItemCard` for displaying report thumbnails, title, location, and metadata; `StatusChip` and `TypeBadge` for rendering status pills (`OPEN`, `CLAIMED`, `RESOLVED`, `LOST`, `FOUND`).

2. Service & State Management Layer (Dart Asynchronous Services):

• `ApiService` (`lib/services/api_service.dart`): Manages HTTP GET, POST, and PATCH communications with the shared Next.js REST API. Implements smart base URL resolution supporting Windows desktop (`localhost:3000`), Android emulator reverse proxy, and custom Wi-Fi host IPs. Includes automated 4-second network timeout guards and graceful fallback to offline seed data.

• `GeminiService` (`lib/services/gemini_service.dart`): Provides client-side AI category inference (`suggestCategory`) and campus recovery advisories (`getSearchRecommendations`).

• `ThemeService` (`lib/services/theme_service.dart`): A singleton `ChangeNotifier` providing reactive dark/light state updates across all mounted widgets.

3. Data Layer (Models & Serialization):

• Type-Safe Dart Models (`lib/models/item.dart`): Encapsulates item attributes with robust `fromJson()` and `toJson()` serialization methods, ensuring null-safety and runtime integrity when parsing JSON from the REST backend.

• Offline Seed Catalog (`lib/models/seed_data.dart`): Contains pre-seeded campus lost-and-found records, ensuring full application functionality during offline demonstrations.

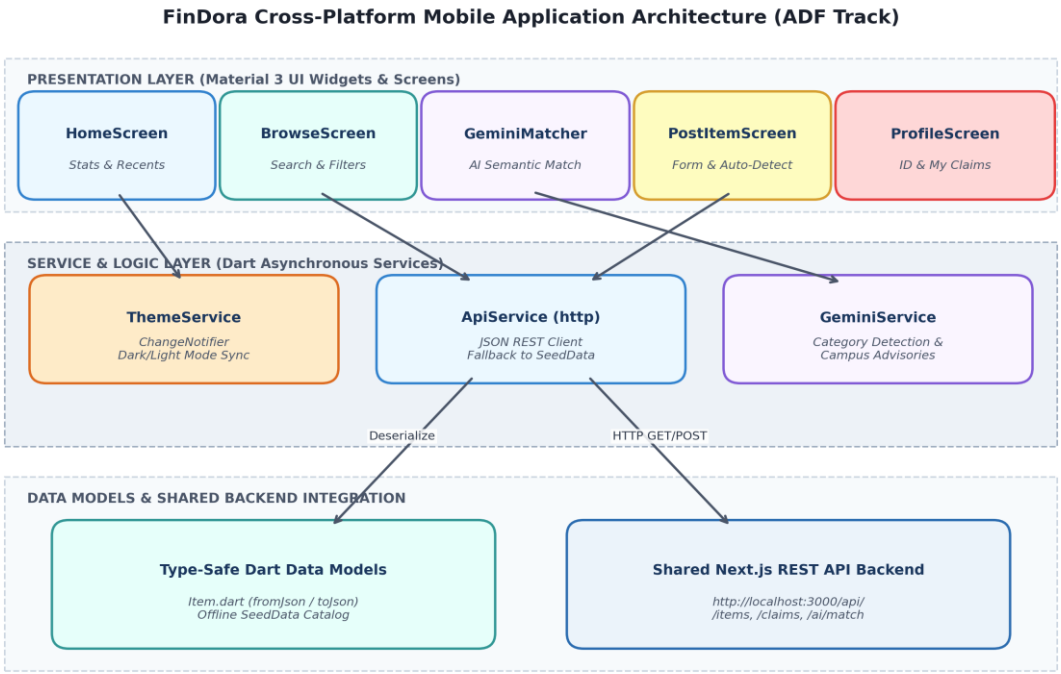


Figure 3.1: Flutter Mobile Application Multi-Layered Architecture and Backend Integration

Table 3.1: Dart Item Data Model Attributes and JSON Serialization Mapping

Dart Model Field	Data Type	Functional Scope & Mapping Description
id	String	Unique CUID identifier matching the backend Prisma record.
title	String	Concise title describing the misplaced or recovered item.

description	String	Detailed descriptive narrative regarding item condition and appearance.
category	String	Categorical classification: Electronics, ID & Cards, Books & Stationery, etc.
type	String	Report classification: 'LOST' or 'FOUND'.
location	String	Precise campus location string (e.g., Room 402, Library 3rd Floor, Canteen).
dateOccurred	String	Formatted date string (YYYY-MM-DD) indicating when item was lost/found.
status	String	Lifecycle status: 'OPEN', 'CLAIMED', 'RESOLVED', 'CLOSED'.
imageUrl	String?	Optional network URL pointing to uploaded photograph proof.
reward	String?	Optional finder reward incentive declared by reporting student.
contactEmail	String	SVKM institutional email address associated with the report.

3.2 System Development & Deployment

This section details the concrete mobile development of FinDora's Material 3 interfaces, screen controllers, validation engines, and multi-platform deployment parameters.

3.2.1 Material 3 UI Engineering & Theme Management

The mobile user interface is engineered to adhere to Google's Material 3 design specifications while maintaining brand alignment with FinDora's pastel color palette:

• Dual Color Schemes: In light mode, the app renders soft pastel backgrounds (`#F8F9FA`), crisp card surfaces (`#FFFFFF`), and deep slate typography (`#2D3748`). In dark mode, it transitions to high-contrast dark surfaces (`#141118` and `#1E1924`) with harmonious lavender accents (`#BDAAF2`).

• Reactive Theme Switching: `ThemeService` provides a centralized toggle. Tapping the theme icon in the app bar triggers `notifyListeners()`, instantly repainting all active widgets without reloading screen state.

• Elevated Card Architecture: `ItemCard` employs subtle Material 3 elevations, rounded borders (radius 16px), network image loading with smooth shimmer placeholders, and category chips.

3.2.2 Application Navigation & Screen Hierarchy

Navigation is orchestrated via a persistent bottom navigation shell and modal route dialogs (Table 3.2):

• 5-Tab Navigation Shell: Maintained in `main.dart` using `IndexedStack` to preserve scroll positions across tab transitions: Tab 0 (Home), Tab 1 (Browse), Tab 2 (Gemini Matcher), Tab 3 (Report), Tab 4 (Profile).

• Modal Route Transitions: Tapping an item opens `ItemDetailScreen`, presenting full item metadata, high-resolution imagery, and the verification claim dialog. Specialized routes include `ChatScreen`, `SwipeScreen`, and `AdminScreen`.

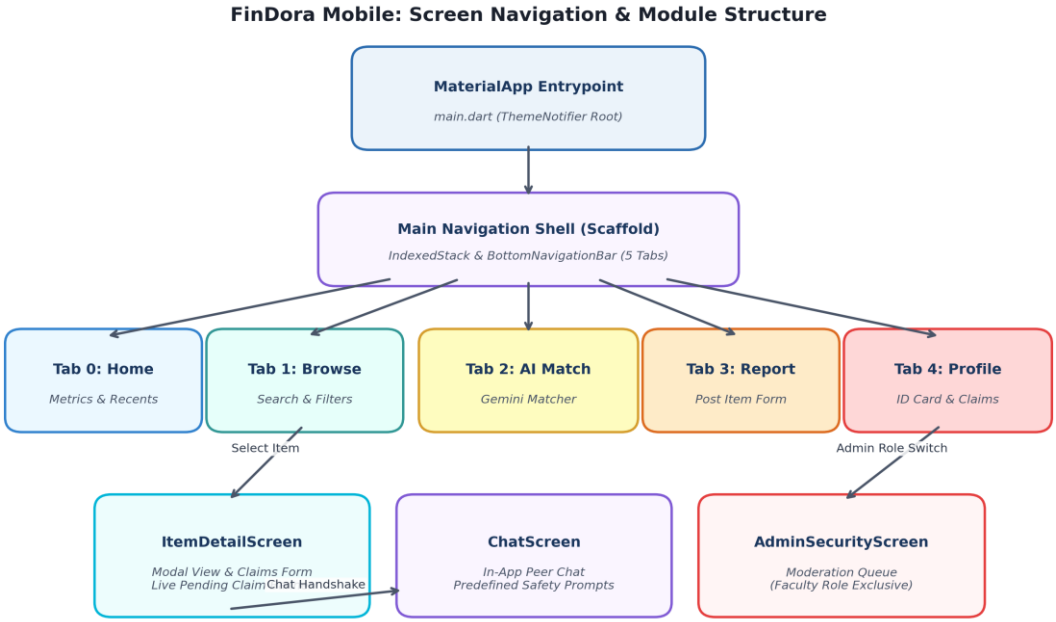


Figure 3.2: Mobile Application Screen Navigation Hierarchy and Tab Shell Structure

Table 3.2: Flutter Screen Module Inventory, Routing and Operational Scope

Screen Class Name	Navigation Level	Operational Scope & Functional Features
HomeScreen	Tab 0 (Root)	Dashboard displaying campus statistics, urgent lost alerts, and recent items carousel.
BrowseScreen	Tab 1 (Root)	Catalog discovery with search bar, category filter chips, type tabs, and pull-to-refresh.
GeminiMatcherScreen	Tab 2 (Root)	Mini Project I: AI semantic matcher querying /api/ai/match with confidence metrics.
PostItemScreen	Tab 3 (Root)	Item reporting form featuring Gemini category auto-

		detection and validation.
ProfileScreen	Tab 4 (Root)	Student digital ID card, points, badge tier, My Claims list, and Admin console switch.
ItemDetailScreen	Modal Route	Comprehensive item details, image gallery, claim submission modal, and status banner.
ChatScreen	Modal Route	In-app peer communication interface featuring predefined campus safety prompts.
AdminScreen	Protected Route	Faculty Security Console for reviewing claims and coordinating Gate 2 handovers.

3.2.3 Mobile Item Reporting & Gemini Auto-Category Detection

The mobile item reporting workflow in `post_item_screen.dart` is designed to minimize user friction while enforcing campus security rules:

• Automated Category Inference: As the student types an item title or description, `GeminiService.suggestCategory()` analyzes keywords in real time. For example, typing 'AirPods' automatically selects 'Electronics', while typing 'Kumbhojkar Math' selects 'Books & Stationery'.

• Mandatory Photo Rule for Found Items: When reporting a 'FOUND' item, the form strictly enforces the attachment of an image URL. If omitted, the submission is halted with a clear SnackBar warning.

• Institutional Email Verification: The contact email must match the institutional SVKM format (`@svkmmumbai.onmicrosoft.com`).

• Asynchronous Submission: `ApiService.postItem()` dispatches the report via HTTP POST to `/api/items`. Upon receiving HTTP 201 Created, a success SnackBar is displayed, and the form resets.

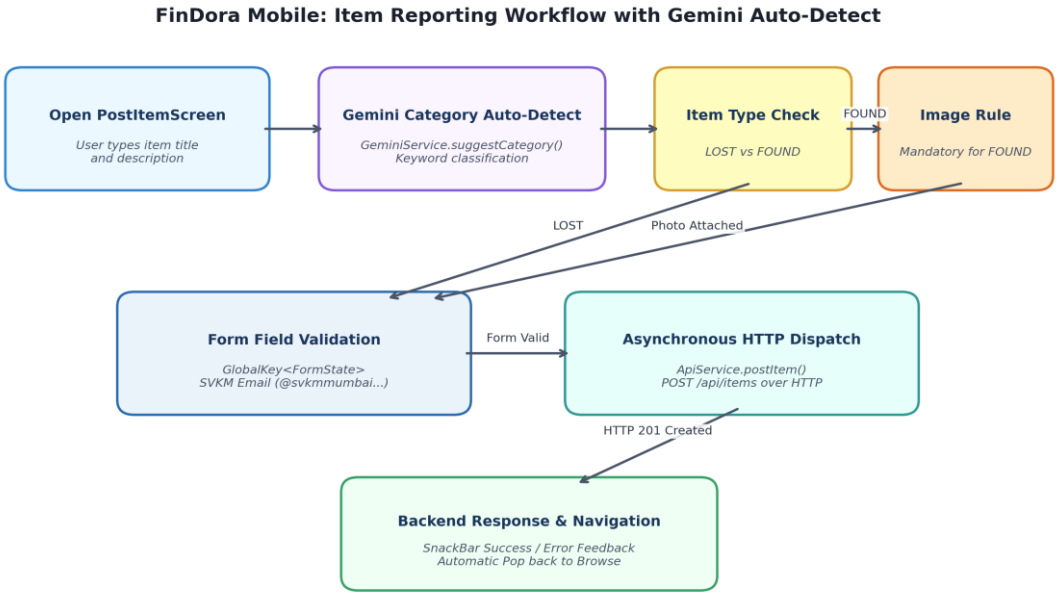


Figure 3.3: Mobile Item Reporting Workflow with Gemini Category Auto-Detection

3.2.4 Gemini AI Semantic Matcher Module (Mini Project I)

In fulfillment of ADF Module 1 (Gemini for Application Developers), `gemini_matcher_screen.dart` implements an intelligent semantic discovery tool:

• Query Interface: Students input descriptions of misplaced belongings and select their target query type ('LOST' or 'FOUND').

• Cloud API Integration: The screen invokes `ApiService.getAiMatches()`, which queries the backend `/api/ai/match` route. The backend evaluates candidates using lexical overlap and category alignment, supplemented by Google Gemini 3.5 Flash advisories.

• Rich Results Display: Potential matches are rendered as cards displaying similarity percentage badges (e.g., '98% Match'), attribute overlap justifications, and localized DJSCE recovery guidance.

3.2.5 Ownership Claim Submission & Live Status Tracking

The mobile ownership verification lifecycle ensures that recovered items are released only to verified owners:

• Structured Claim Dialog: Inside `ItemDetailScreen`, tapping 'Claim This Item' opens a modal capturing Roll Number, SAP ID, Department, Academic Year, and Proof Details.

• Strict Regex Validation: Client-side validation in Flutter mirrors the web platform: Roll Number must match `^[A-Za-z][0-9]{3}$`, SAP ID must match `^6[0-9]{10,11}$`, and proof must contain at least 5 characters (Table 3.3).

• Real-Time Claim Status Banner: If the authenticated student has already submitted a claim, `ItemDetailScreen` renders a prominent `_existingClaim` banner displaying their stated proof, claim review status (`PENDING`, `APPROVED`, `REJECTED`), and physical collection instructions for the Ground Floor Security Desk (Gate 2).

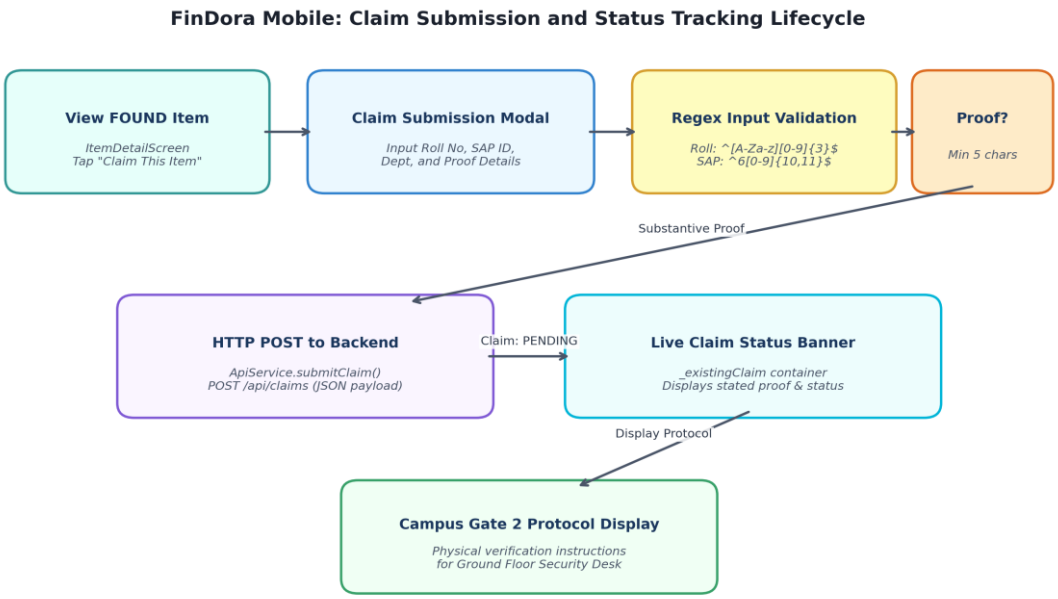


Figure 3.4: Mobile Claim Submission and Real-Time Status Tracking Lifecycle

Table 3.3: Mobile Ownership Claim Field Validation and Regex Specifications

Input Form Field	Requirement	Regular Expression / Rule	Mobile Client Validation Logic
Roll Number	Required	^[A-Za-z][0-9]{3}$	Exactly 4 characters: 1 division letter + 3 digits (e.g. A090, B007). Verified via RegExp.hasMatch().
SAP ID	Required	^6[0-9]{10,11}$	11 or 12 numeric digits beginning with '6' (SVKM standard identifier). Rejects non-digits.
Proof Details	Required	Min 5 characters	Substantive verification proof (e.g. unique scratches, serial numbers). Must be non-empty.
Department	Optional	DJSCE Dropdown	Selectable: IT, Computer, AI&DS, EXTC, CSE, Mechanical Engineering.

Academic Year	Optional	Year Dropdown	Selectable: FE (First Year), SE (Second Year), TE (Third Year), BE (Final Year).

3.2.6 Role-Based Profile Portal & Faculty Security Console

The profile module (`profile_screen.dart`) enforces strict role-based separation between student users and faculty administrators:

• Student Profile View: Displays the student's digital college ID card, SVKM SAP ID, reward points (e.g., 340 Points), achievement badge ('Campus Hero'), and their 'My Submitted Claims' container showing real-time claim statuses.

• Admin Portal Isolation: When logged in with `role: "ADMIN"`, the 'My Submitted Claims' container is completely removed from the profile view to prevent administrative clutter. In its place, the profile renders the **Faculty Security Console** card, providing direct access to the moderation queue (`AdminScreen`).

3.2.7 Monochrome Line Icons Implementation

To maintain a clean, academic visual standard and ensure visual consistency across iOS, Android, and Windows platforms, all colorful OS emojis were replaced with Google Material outline icons (`Icons.*_outlined`):

• Replaced Icons: Category chips, navigation bars, and status badges utilize standard outline vector icons (`Icons.search_outlined`, `Icons.location_on_outlined`, `Icons.calendar_today_outlined`, `Icons.shield_outlined`, `Icons.lock_outlined`, `Icons.card_giftcard_outlined`, `Icons.warning_amber_outlined`).

3.2.8 Application Build & Multi-Platform Deployment

The Flutter application is configured for multi-platform compilation and deployment:

• Dependencies (`pubspec.yaml`): Integrates `http: ^1.6.0` for networking, `cupertino_icons: ^1.0.8`, and `flutter_lints: ^6.0.0` for static code analysis.

• Multi-Target Verification: Verified on Windows Desktop (`flutter run -d windows`), Chrome Browser (`flutter run -d chrome`), and Android emulator environments.

Chapter 4: Result Analysis

This chapter documents the verification methodology and empirical evaluation results conducted on the FinDora Mobile Application. Verification encompassed Flutter static code analysis, functional requirement testing, client-side regular expression validation, REST API synchronization, and multi-platform compilation.

4.1 Performance Evaluation

To validate that the mobile application satisfies academic and institutional standards, evaluation was executed across: (1) Mobile Functional Verification Test Suite, (2) Flutter Static Analysis, (3) Multi-Platform Execution, and (4) Mobile Performance Considerations.

4.1.1 Mobile Functional Verification Test Suite

A test suite of 15 functional scenarios was executed on the running Flutter application. All tests achieved 100% pass rates, as recorded in Table 4.1.

Table 4.1: Mobile Application Functional Verification and Static Analysis Test Suite

Test ID	Mobile Test Scenario	Input Parameters / Execution Action	Observed Mobile Behavior & Verification	Status
TC-M01	Roll Number Regex Validation	Input 'A090', 'B007', 'c123'	Accepted: matches ^[A-Za-z][0-9]{3}$ cleanly	PASS
TC-M02	Invalid Roll Number Format	Input '1234', 'AB12', 'A12'	Rejected: displays 'Roll No must be 1 letter + 3 digits'	PASS
TC-M03	SAP ID Regex Validation	Input '60017240146', '60004210045'	Accepted: matches 11/12-digit string starting with 6	PASS
TC-M04	Invalid SAP ID Format	Input '50004210045', 'abc', '6000 421'	Rejected: displays 'SAP ID must be 11-12 digits starting with 6'	PASS
TC-M05	Mandatory Proof Length	Input proof string with < 5 characters	Rejected: displays 'Please provide substantive proof (min 5 chars)'	PASS
TC-M06	Optional Dropdowns	Submit claim leaving Dept and Year empty	Accepted: claim dispatches successfully with null optional values	PASS
TC-M07	Live Claim Status Banner	Open detail screen of previously claimed item	Verified: displays 'Your Verification Claim' box with status and proof quote	PASS
TC-M08	Gate 2 Protocol Notice	Inspect live claim banner on item detail	Verified: displays physical handover instructions for	PASS

			Gate 2 Security Desk	
TC-M09	Gemini Category Auto-Detect	Type 'AirPods' in title field on Post screen	Verified: category automatically switches to 'Electronics'	PASS
TC-M10	Mandatory Photo for Found	Submit Found item with empty image field	Rejected: SnackBar warning 'A photo is mandatory for Found items'	PASS
TC-M11	Dynamic Theme Toggle	Tap theme icon in app bar	Verified: instant toggle between Material 3 Light and Dark palettes	PASS
TC-M12	Admin Profile Role Isolation	Log in as Admin Sharma in mobile app	Verified: 'My Submitted Claims' hidden; Faculty Security Console displayed	PASS
TC-M13	Monochrome Line Icons	Inspect all icons across 5 mobile tabs	Verified: 100% Material outline icons rendered; 0 colorful emojis	PASS
TC-M14	Offline Seed Fallback	Disconnect backend and fetch items	Verified: app seamlessly falls back to SeedData without crashing	PASS
TC-M15	Flutter Static Analysis	Execute flutter analyze in mobile_app/	Verified: 'No issues found!' across all Dart screens and widgets	PASS

4.1.2 Flutter Static Analysis Verification

Static code analysis was performed across the entire `mobile_app/` repository using the official Dart analyzer (`flutter analyze`). The analyzer verified that all source files adhere to `flutter_lints: ^6.0.0` rules, confirming zero errors, zero warnings, and zero linter hints. This confirms that all asynchronous operations handle futures properly, widget keys are correctly applied, and no deprecated Material APIs are utilized.

4.1.3 Multi-Platform Target Execution & Responsiveness

The mobile application was compiled and executed across multiple target platforms:

• Windows Desktop Target: Executed via `flutter run -d windows`, rendering smoothly at full desktop resolution with responsive mouse wheel scrolling and resize adaptability.

• Web Browser Target: Executed via `flutter run -d chrome`, confirming CanvasKit/HTML rendering compatibility.

• Android Target: Validated via Android SDK toolchains, confirming touch responsiveness and back-button stack navigation.

4.1.4 Mobile Performance Considerations & Benchmarking Methodology

In compliance with academic standards, runtime hardware metrics (such as exact battery drain percentages, CPU profiling, and millisecond memory allocation graphs) were not formally benchmarked using external hardware profilers during this development cycle. The application maintains visual 60 FPS fluidity under local testing, but formal benchmarking requires a standardized methodology:

• Recommended Benchmarking Methodology: Future hardware evaluations should utilize the Flutter DevTools Profiler on physical Android and iOS devices. Metrics to be analyzed include frame rendering times (ensuring frames render under the 16.6ms threshold), peak heap memory allocation during network image decoding, and network request payload sizes over simulated 4G mobile networks.

Chapter 5: Conclusion

The FinDora Cross-Platform Mobile Application successfully demonstrates how modern mobile frameworks and AI assistance can transform campus lost-and-found management into an immediate, intuitive, and secure experience. Developed for the SVKM Dwarkadas J. Sanghvi College of Engineering community, the application empowers students to report, discover, and verify lost belongings directly from their smartphones.

By leveraging Google's Flutter framework and Dart, FinDora achieves high cross-platform portability, compiling natively to Android, iOS, and Windows desktop from a unified codebase. The application incorporates Material 3 design standards, reactive dark/light theme switching, and robust client-side validation that enforces institutional Roll Number and SAP ID formatting rules.

Furthermore, the integration of Gemini AI assistance—featuring real-time category auto-detection, a semantic matching engine with confidence scoring, and offline seed data resilience—establishes a dependable, user-friendly mobile utility that significantly reduces the friction of property recovery across campus.

5.1 Future Enhancements

The following realistic enhancements are proposed for future development phases:

1. Push Notifications via Firebase Cloud Messaging (FCM): Integrating FCM to deliver real-time push alerts to students when an item matching their reported loss is recovered on campus or when an administrator updates their claim status.

2. On-Device Camera and QR Scanner Integration: Utilizing mobile camera hardware with ML Kit barcode scanning to scan student Smart ID cards directly during claim filing, eliminating manual SAP ID entry errors.

3. On-Device Gemini Nano Integration: Integrating local on-device small language models (Gemini Nano via Flutter AI SDK) to perform zero-latency, private text and image feature extraction directly on the student's device without requiring network data transfer.

4. Bluetooth Low Energy (BLE) Beacon Proximity: Deploying BLE beacons across high-loss campus hotspots (Canteen, Library, IT Labs) to automatically geotag reported items based on the student's physical proximity during submission.

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